
Formal Specification Logic: Linear Temporal Logic (LTL)

Modal Logics



- Tense (Modal) logic is a system
 - developed by philosophers for expressing statements about changes in time.
- Temporal logic is a special type of modal logic
 - originally developed by philosophers to study different modes of "truth"
- Temporal logic provides a formal system
 - qualitatively describing and reasoning about how the truth values of assertions change over **time**
 - It is appropriate for describing the time-varying behavior of systems (or programs).

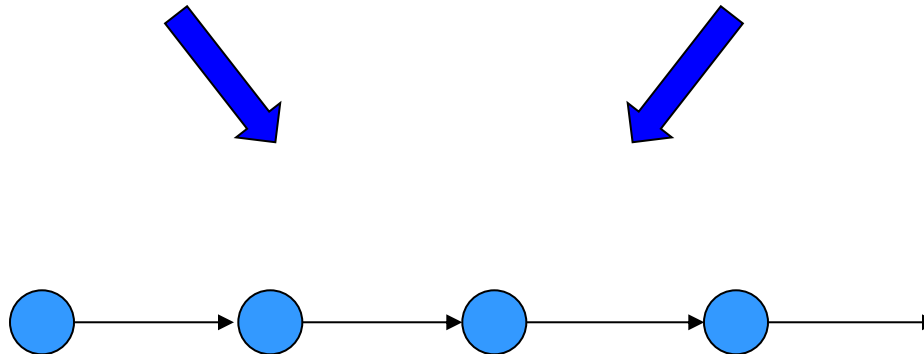
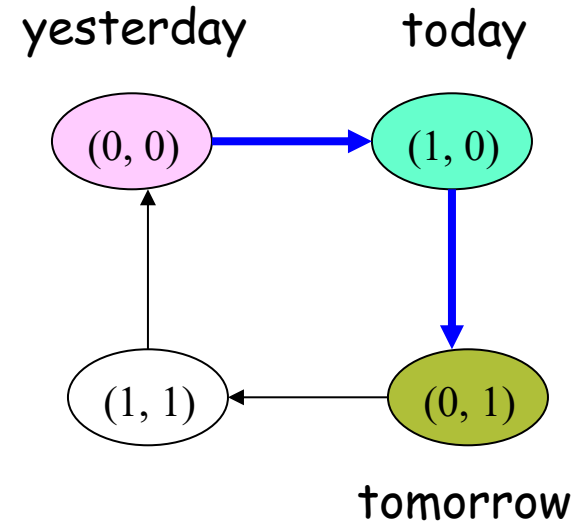
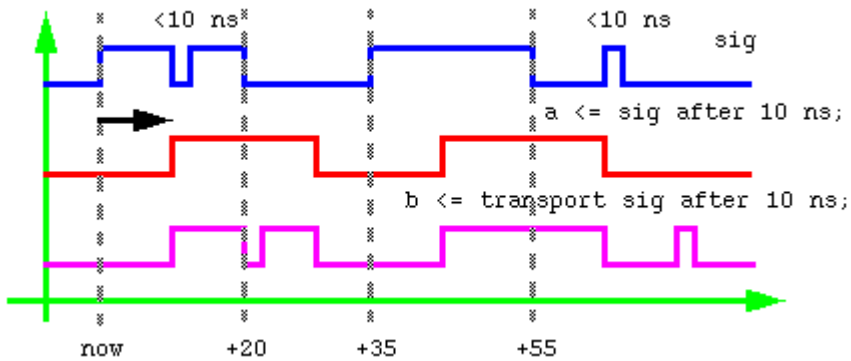
What is Time?



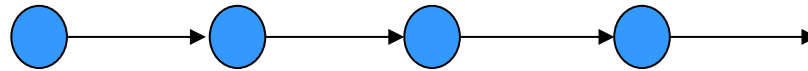
Albert Einstein said:

"...for us physicists believe the separation between **past**, **present**, and **future** is only an illusion, although a convincing one."

Time = Trace

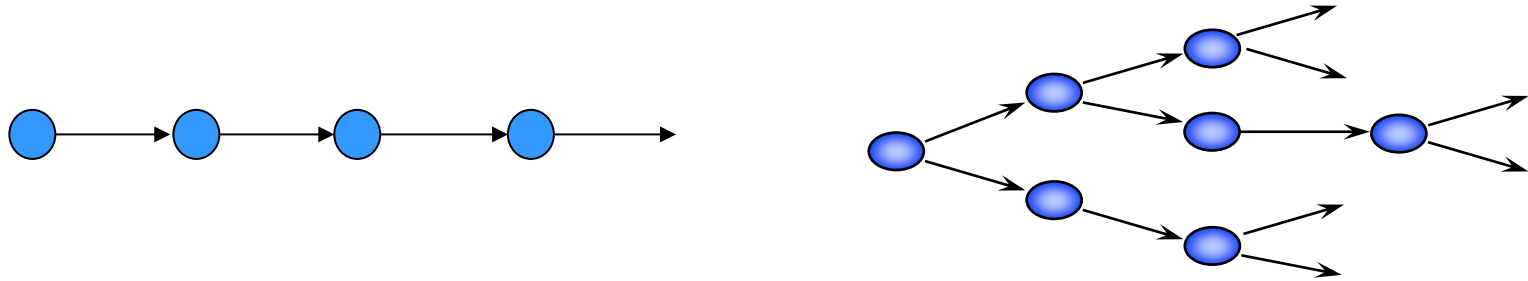


Traces



- A **trace** is a (finite or infinite) **sequence of states**:
 - $\sigma = s_0, s_1, s_2, \dots$
- Given two states s and t , $s < t$ means that **s occurs before t** .
- Time lines: Underlying structure of time
 - a totally ordered set $(S, <)$, isomorphic to $(\mathbb{N}, <)$
 - Discrete
 - an initial moment without predecessors, infinite into the future.
- Each state assigns values to signals/variables.
 - Hardware simulation → clocked traces
 - Formal verification → infinite traces
 - Runtime checking → finite prefixes of traces

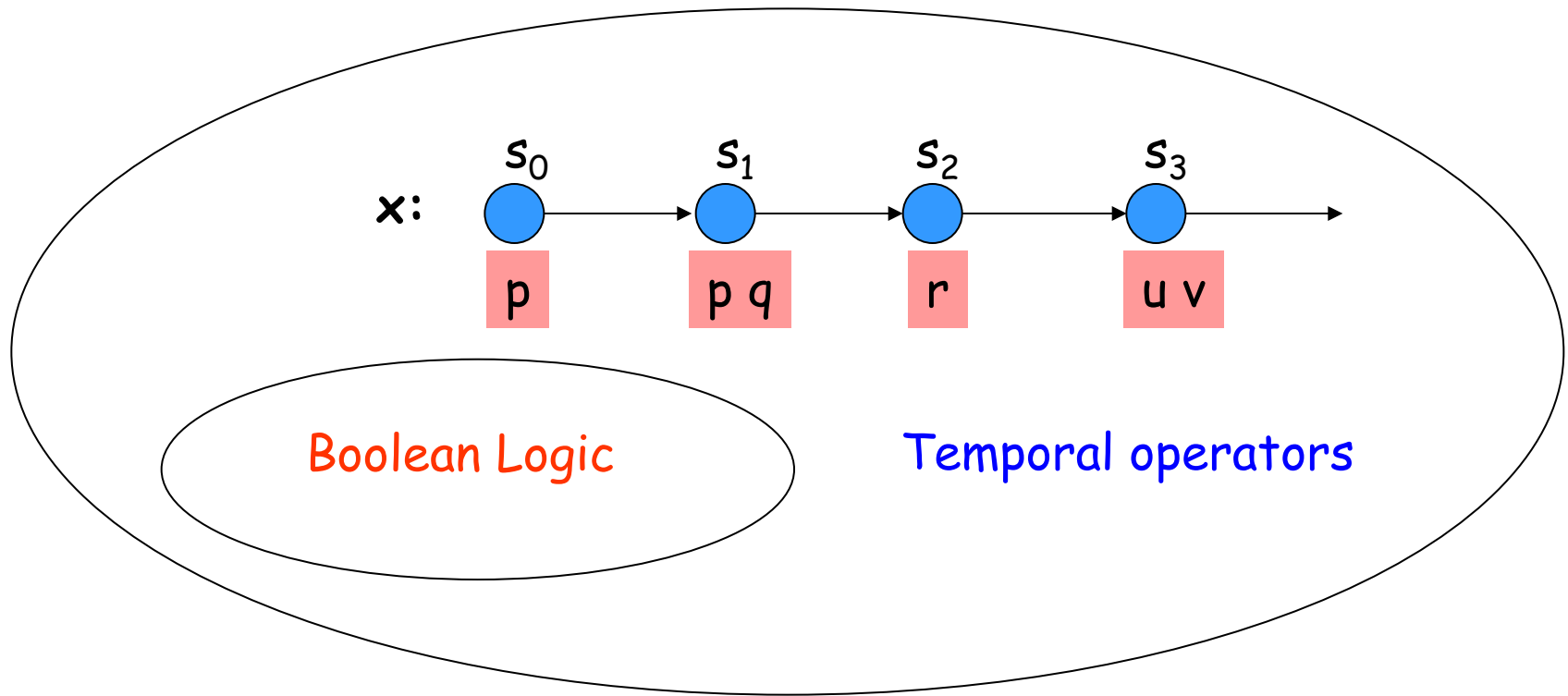
Classification of Temporal Logics



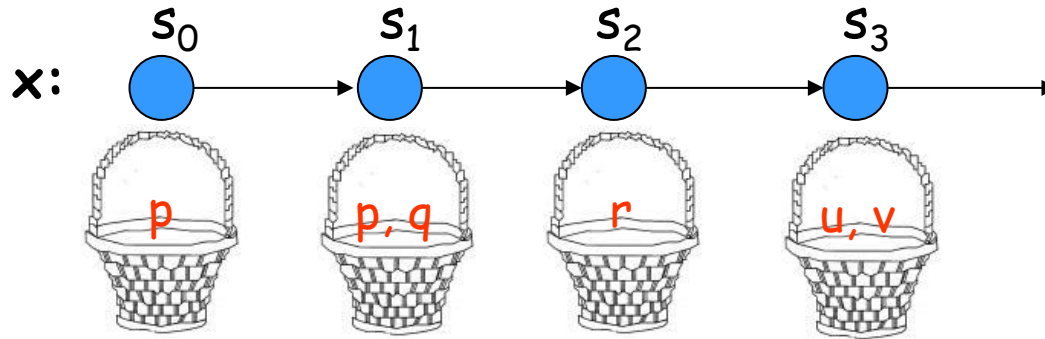
- The underlying nature of time:
 - **Linear**: at each moment there is only one possible future moment
 - **Branching**: at each moment, time may split into alternate courses representing different possible futures
- Other considerations:
 - Propositional versus first-order
 - Point versus intervals
 - Discrete time versus continuous time
 - Past versus future

Propositional Linear Temporal Logic (LTL)

- LTL = Classical propositional logic + time line + temporal operators

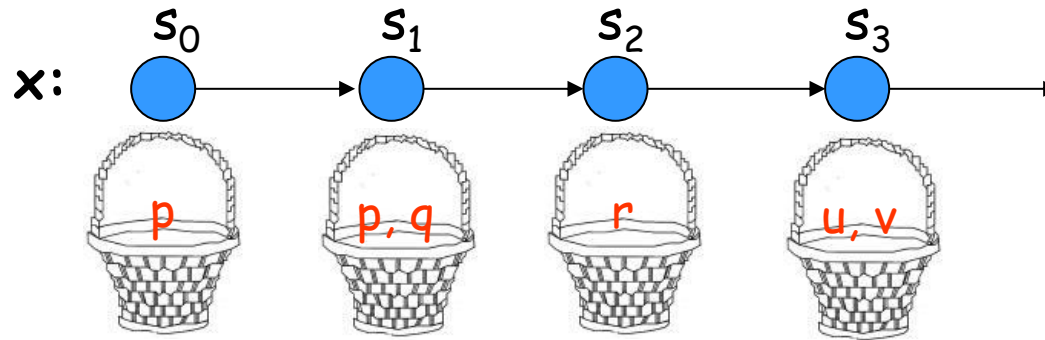


Linear Temporal Logic



- Introduced by Pnueli in 1977
- Every logic needs a model theory.
 - A collection of mathematical structures, and an interpretation that tells us whether a given formula is true in a given structure.

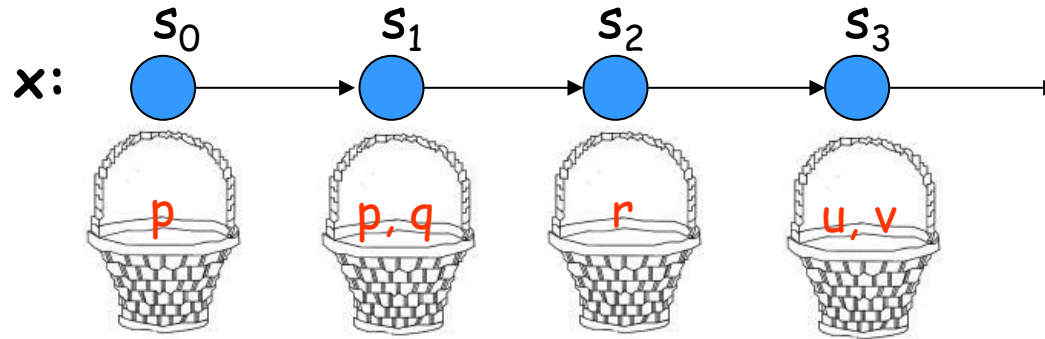
Modeling Infinite Traces



- Linear time structure

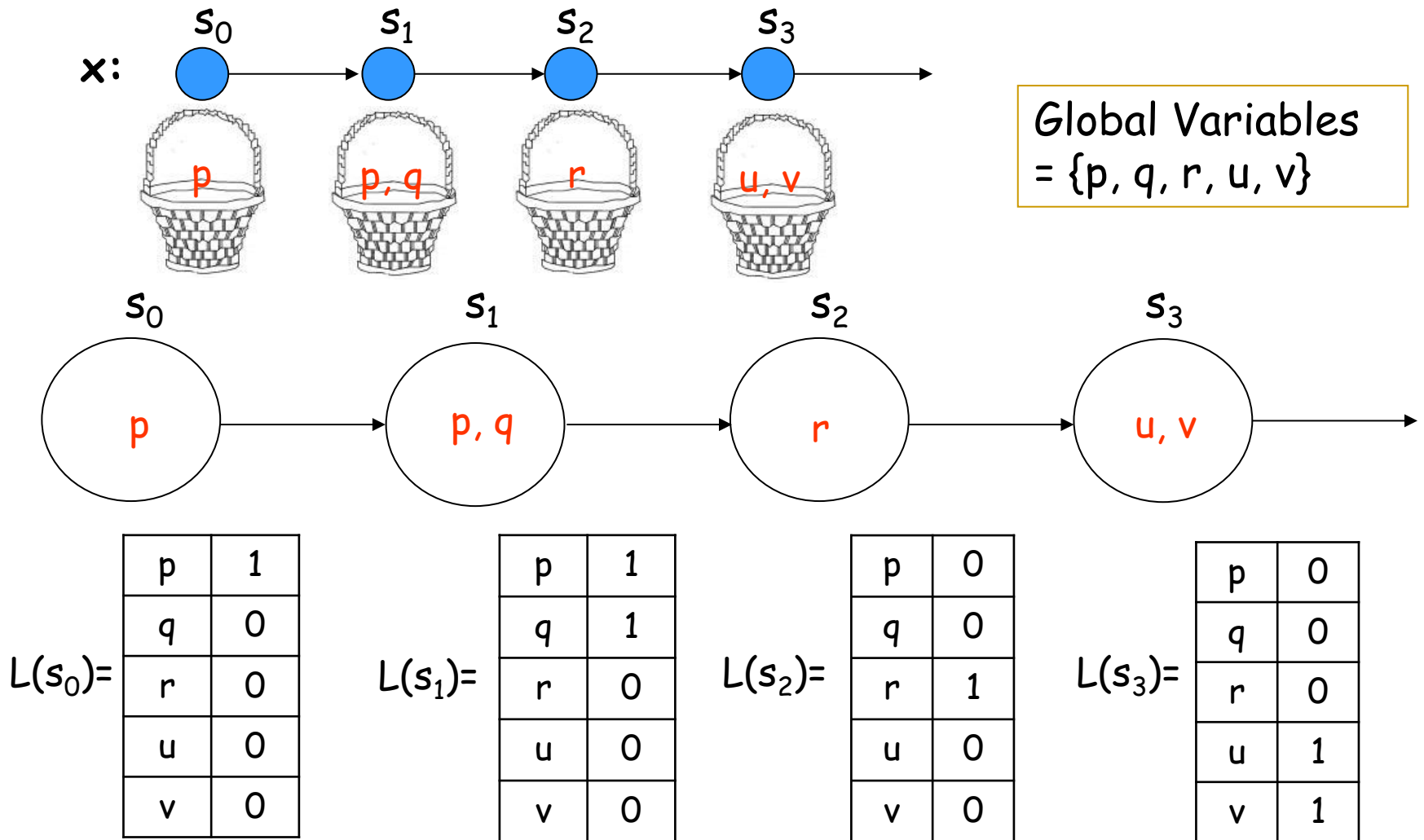
- **Infinite Trace** $x: \mathbb{N} \rightarrow S$ an infinite sequence of states ($x = s_0, s_1, \dots$)

Linear Structure: $B=(S, x, L)$

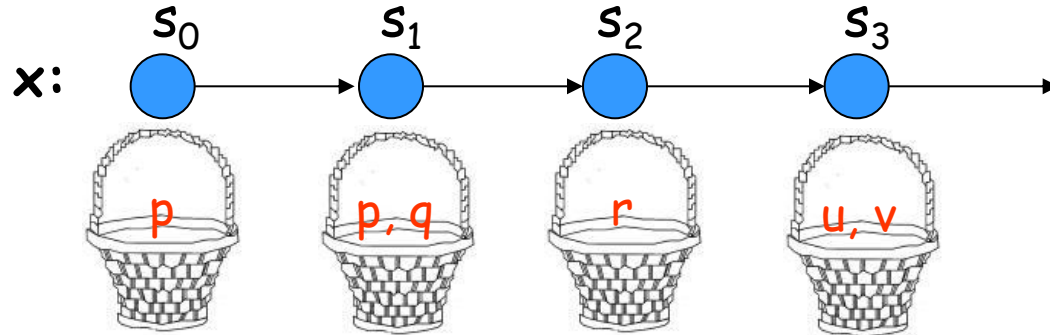


- Linear time structure $B=(S, x, L)$
 - S : a set of abstract states, nodes, moments.
 - $x: \mathbb{N} \rightarrow S$ an infinite sequence of states ($x = s_0, s_1, \dots$),.
 - $L: S \rightarrow 2^{AP}$ labeling each state with the set of atomic propositions in AP true at the state.

L: Local Truth Table for Variables



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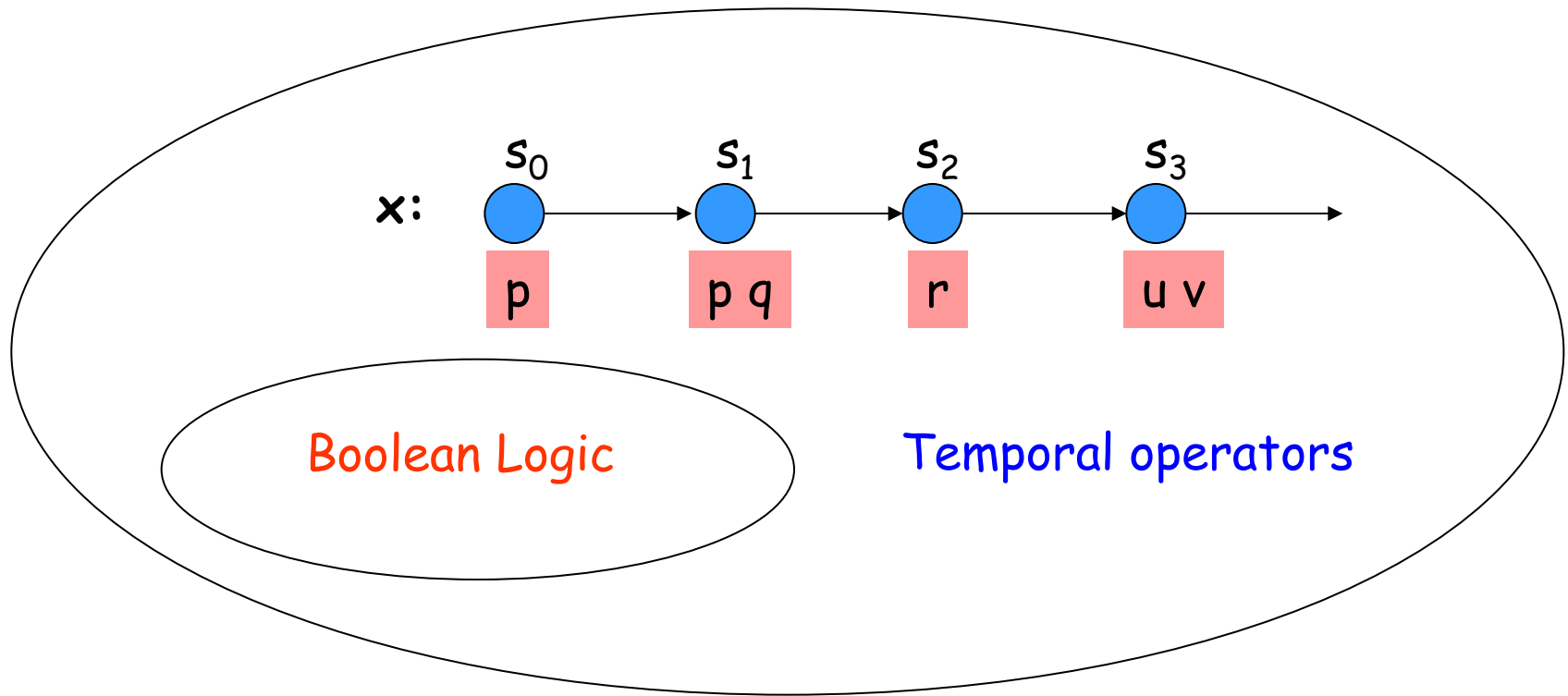
$$\begin{aligned} L(s_0) &= \{p\}, \\ L(s_1) &= \{p, q\}, \\ L(s_2) &= \{r\}, \\ L(s_3) &= \{u, v\}, \\ L: S &\rightarrow 2^{AP} \end{aligned}$$

$AP = \text{Atomic Propositions} = \{p, q, r, u, v\}$

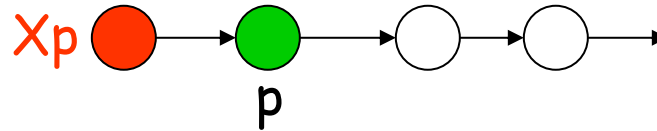
- Linear time structure $B = (S, x, L)$
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Propositional Linear Temporal Logic (LTL)

- LTL = Classical propositional logic + time line + temporal operators

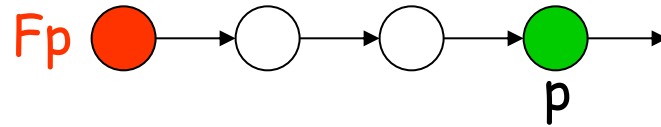


Temporal Operator: X



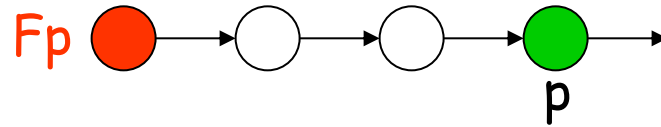
- p = "it rains"
- $X p$ = "it rains tomorrow"

Temporal Operator: F



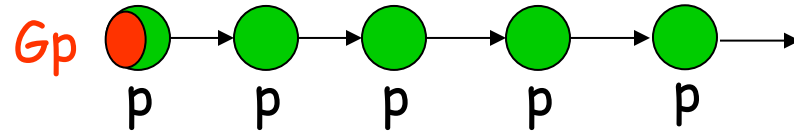
- p = "it rains"
- Fp = "sometimes it rains", "eventually it rains"

Reference Moment: s_0



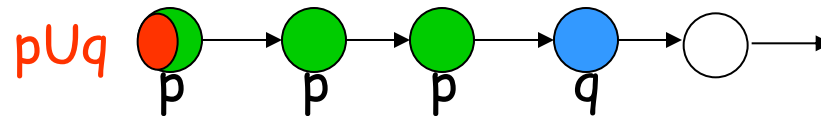
- Formulas are evaluated for truth values at the reference moment s_0 .
- The evaluation is based on the facts on the future moments in respect to s_0 .

Temporal Operator: G



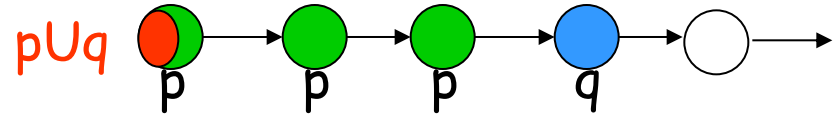
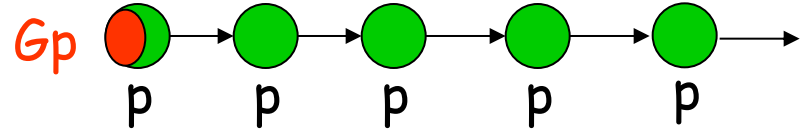
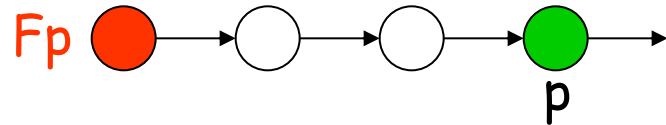
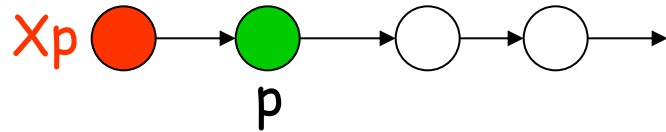
- p = "it rains"
- $G p$ = "it **always** rains"

Temporal Operator: U



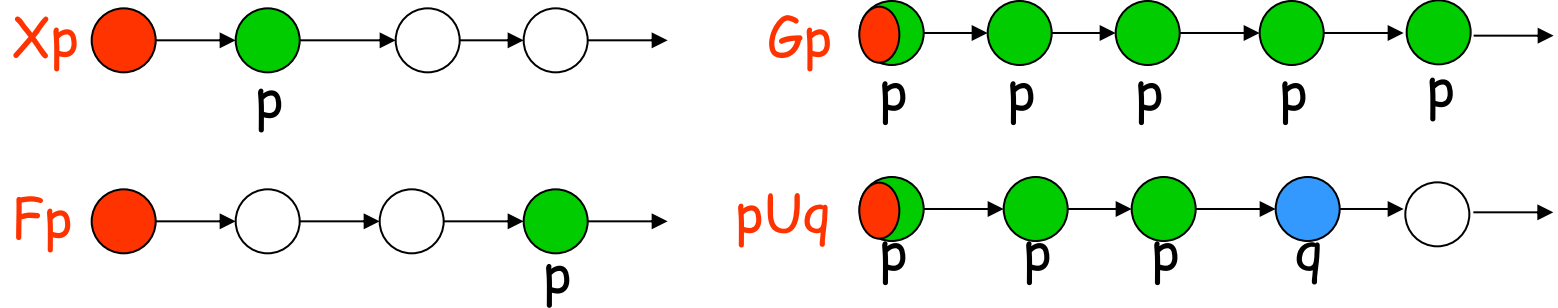
- pUq (Until, “p until q”)

Temporal Operators



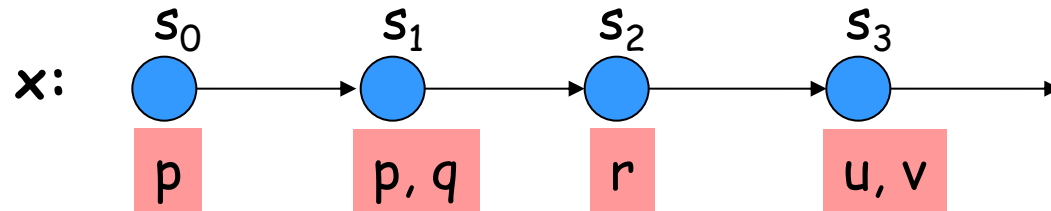
- p = "it rains"
 - $X p$ = "it rains **tomorrow**"
 - $F p$ = "**sometime** it rains", "**eventually** it rains"
 - $G p$ = "it **always** rains"

Temporal Operators



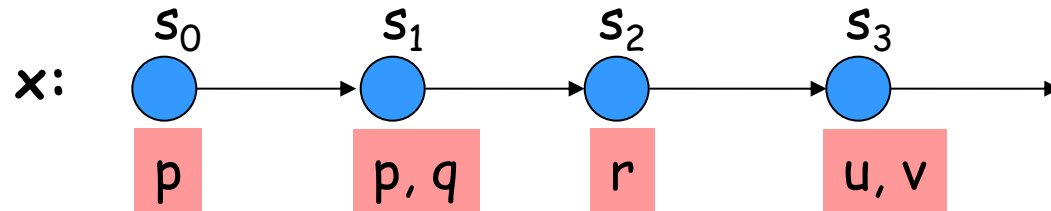
- Basic temporal operators:
 - $F p$ (Future, Finally, $\Diamond p$, "sometime p", "eventually p")
 - $G p$ (Globally, "always p", "henceforth p")
 - $X p$ (Next, Op, "next time p")
 - pUq (Until, "p until q")

Propositional Linear Temporal Logic (LTL)



- Boolean Logic
 - $p = \text{"Grass is green"}$
 - $q = \text{"It rains"}$
 - $r = \text{"The office is closed"}$
 - $u = \text{"x=2"}$
 - $v = \text{"We go to LA"}$

Propositional Linear Temporal Logic (LTL)



■ Boolean Logic

- p = "Grass is green"
- q = "It rains"
- r = "The office is closed"
- u = " $x=2$ "
- v = "We go to LA"

■ LTL

- Fp = "Grass will be eventually green"
- Xq = "It will rain tomorrow"
- Gr = "The office is closed everyday"
- Fu = " $x=2$ is eventually true"
- Fv = "We will go to LA sometime"

Syntax of LTL

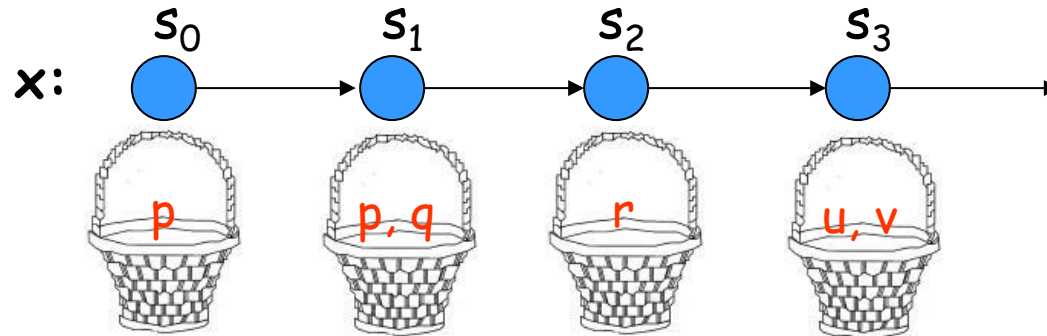
- The set of formulas of PLTL is the least set of formulas generated by the following rules:
 - (1) Atomic propositions are formulas,
 - (2) p and q formulas: $p \wedge q$, $\neg p$, $p \cup q$, and Xp are formulas.

Syntax of LTL

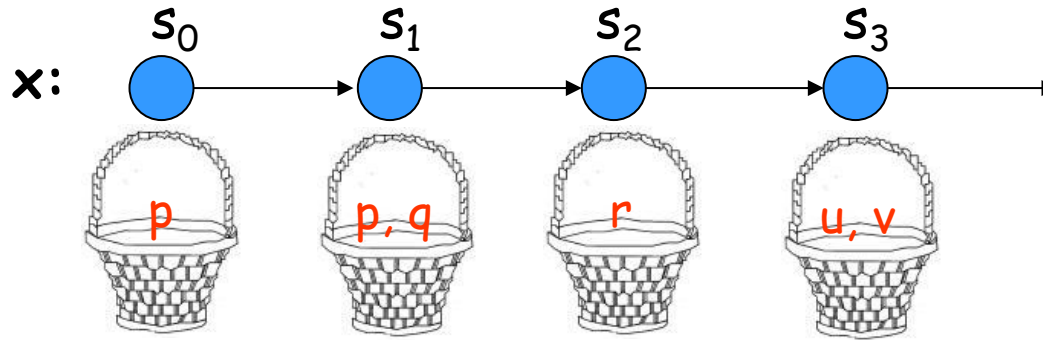
- The other formulas can then be introduced as abbreviations:
 - $p \vee q$ abbreviates $\neg(\neg p \wedge \neg q)$,
 - $p \rightarrow q$ abbreviates $\neg p \vee q$,
 - $p \leftrightarrow q$ abbreviates $(p \rightarrow q) \wedge (q \rightarrow p)$,
 - true abbreviates $p \vee \neg p$,
 - false abbreviates $\neg \text{true}$,
 - Fp abbreviates $(\text{true} \cup p)$,
 - Gp abbreviates $\neg F\neg p$.

Semantic Interpretation

- Semantic interpretation structure $B = (S, x, L)$:
 - **Interpretation** = (B, s_0)

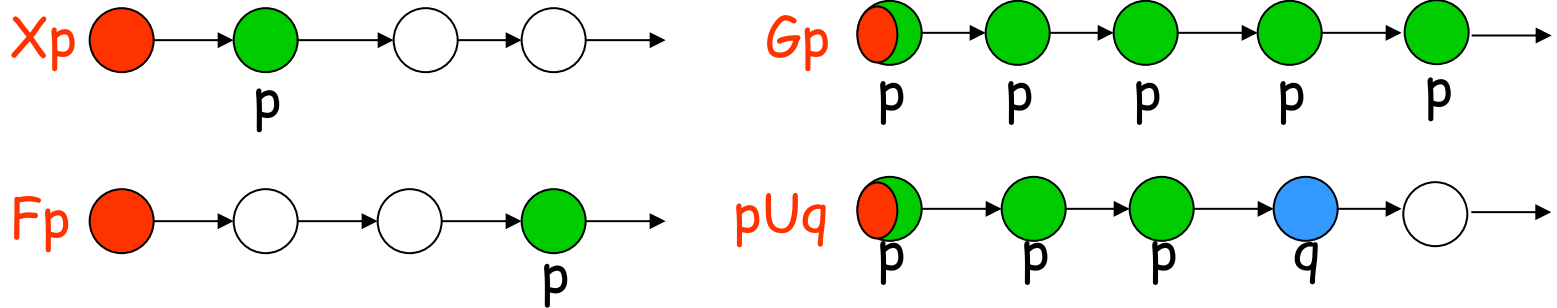


Interpretation = (B, s_0)



- $(B, s_0) \models H$
 - iff "formula H is true at s_0 of timeline x in B ."
- H is *satisfiable*
 - iff there exists $B=(S, x, L)$ such that $B, s_0 \models H$.
- Any such structure defines a *model* of H .

Semantics of LTL



■ Semantics

$B, s_0 \models p$ iff $p \in L(s_0)$, for atomic proposition p

$B, s_0 \models p \wedge q$ iff $B, s_0 \models p$ and $B, s_0 \models q$

$B, s_0 \models \neg p$ iff it is not the case that $B, s_0 \models p$

$B, s_0 \models p \cup q$ iff $\exists j. (s_j \models q \text{ and } \forall k < j (s_k \models p))$

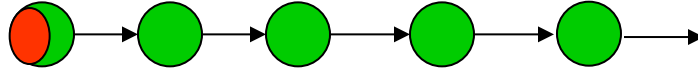
$B, s_0 \models Xp$ iff $s_1 \models p$

$B, s_0 \models Fp$ iff $\exists j. (s_j \models p)$

$B, s_0 \models Gp$ iff $\forall j. (s_j \models p)$

$k, j = 0, 1, 2, \dots$

Present & Future

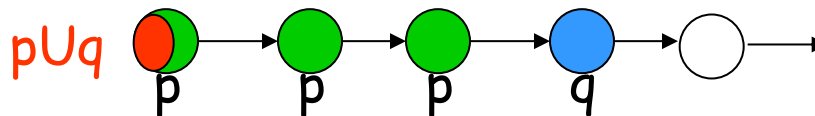
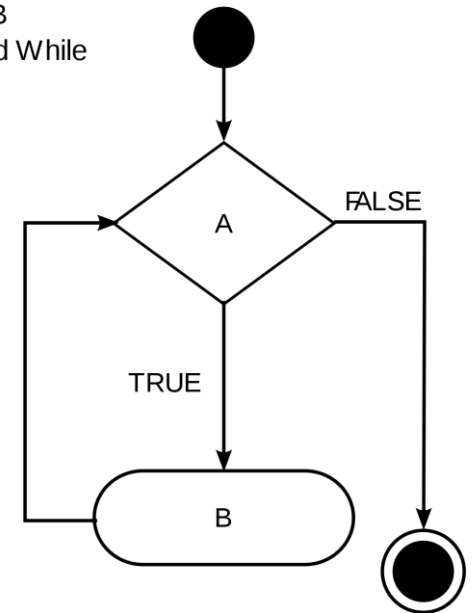


- In our definitions, the future includes the present.
- Accordingly, “in all future states” is understood to include the present state.
- It is a matter of convention.

$$p \cup q$$

- Loop while statement
- Execution trace:
 - $A=1, B, A=1, B, A=1, B, A=1, B, A=1, B, A=0$ end
- Let $p = \langle A=1, B \rangle$ and $q = (A=0)$
- Execution trace: p, p, p, p, p, q

While (A= TRUE) Do
 B
 End While



Weak "Until" Operator

- The "weak" until:

$$p \ W \ q = p \ U \ q \vee Gp$$

- The weak until allows the possibility that q never happens and p remains true forever.

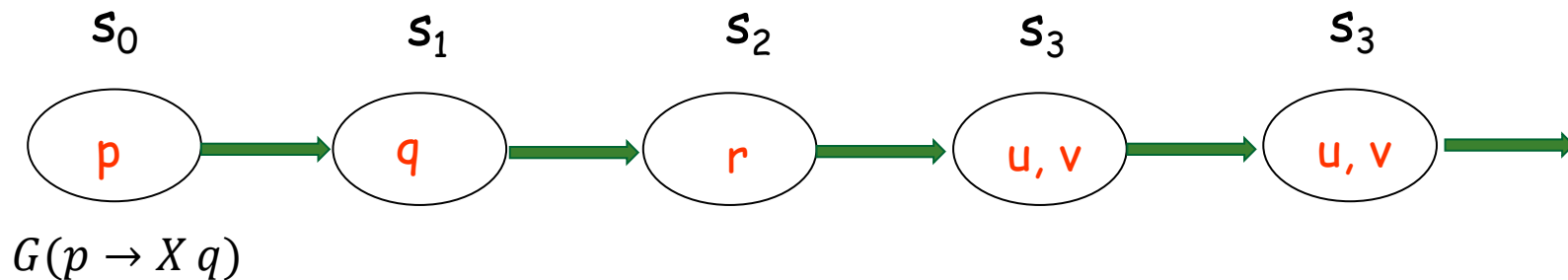
- Useful for expressing properties like:

$$\neg \text{output}(x) \ W \ \text{input}(x)$$

- "An output does not occur before an input occurs."

Example: $G(p \rightarrow X q)$

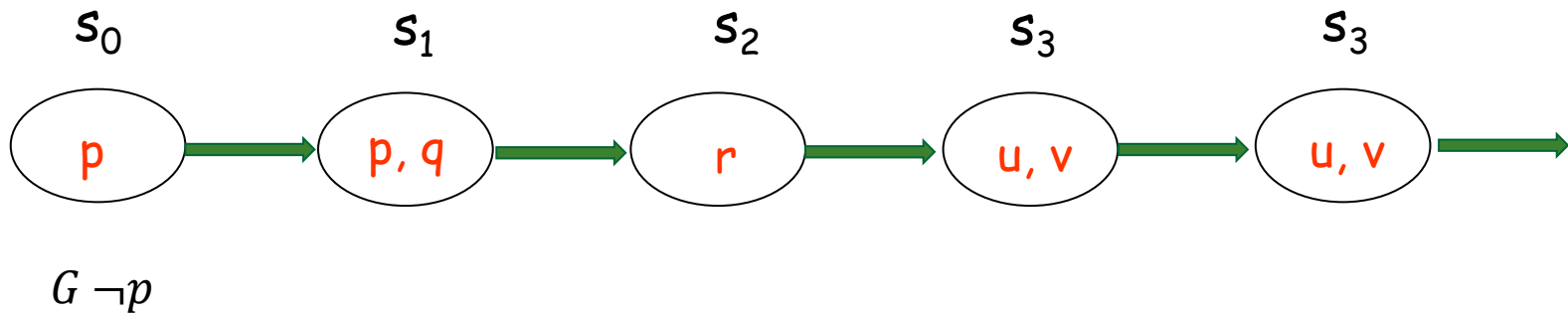
- “If p ever happens, q must happen immediately next.”
 - $G(p \rightarrow X q)$



- $B, s_0 \models G(p \rightarrow X q)$
- $B, s_1 \models G(p \rightarrow X q)$
- $B, s_3 \models G(p \rightarrow X q)$

Example: $G \neg p$

- “p never happens.”
- $G \neg p$



- $B, s_0 \not\models G \neg p$
- $B, s_1 \not\models G \neg p$
- $B, s_3 \models G \neg p$

Finitely Often $\rightarrow$ Infinitely Often

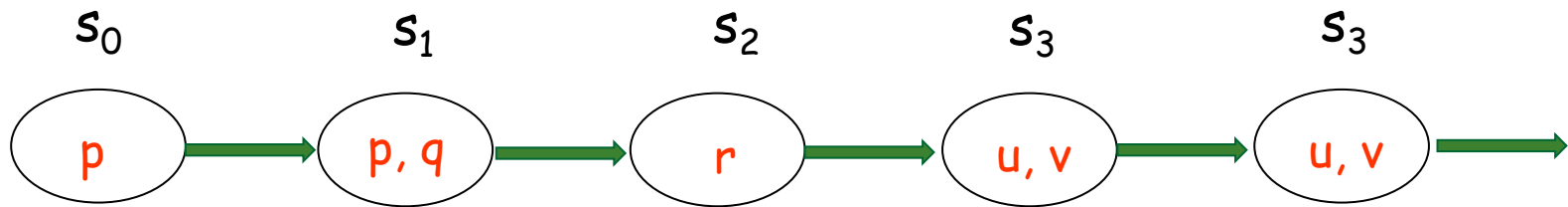
- Consider an event (or state) p along an **infinite run**:
- **Finitely often**
 - p happens only a finite number of times, then stops forever.
- **Infinitely often**
 - p keeps happening again and again, never stopping.

Infinitely Often in LTL

- **Infinitely often:**
 - GFp
- **Finitely often:**
 - $FG\neg p$
 - Eventually, p stops forever
- They are **logical opposites** on infinite traces.

Example: $GF\ q$

- “q happens infinitely often.” (“globally, eventually q”)
- $GF\ q$

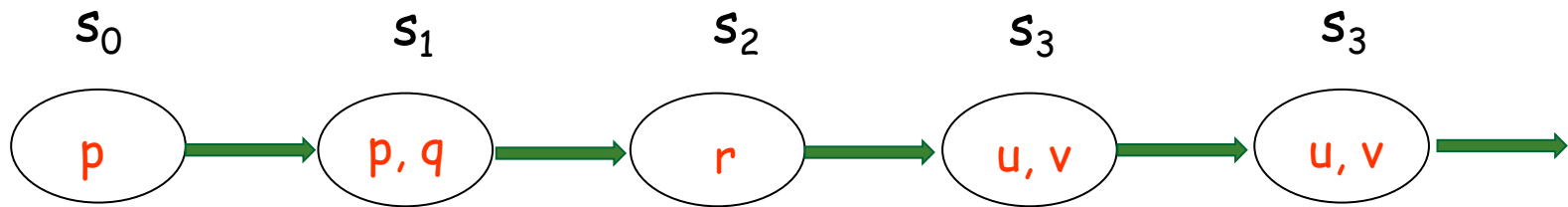


$GF\ q$

- $B, s_0 \models ?$
- $B, s_1 \models ?$
- Describe the finite patterns in an infinite trace.

Example: $G(r \rightarrow (p U q))$

- "Whenever r happens, p must be true until q holds."
- $G(r \rightarrow (p U q))$

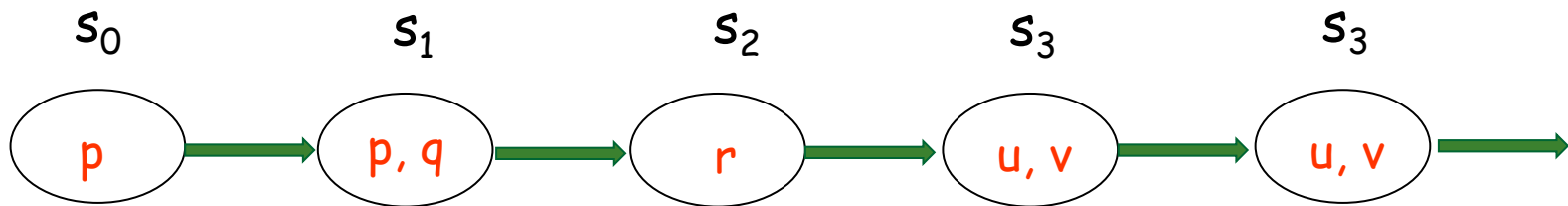


$G(r \rightarrow (p U q))$

- $B, s_0 \models ?$
- $B, s_1 \models ?$
- $B, s_3 \models ?$

Example: $Fr \wedge G(r \rightarrow XG\neg r)$

- “r happens at least once and then never again.” (Once r occurs, the next state must begin a forever-no-r region.)
- $Fr \wedge G(r \rightarrow XG\neg r)$



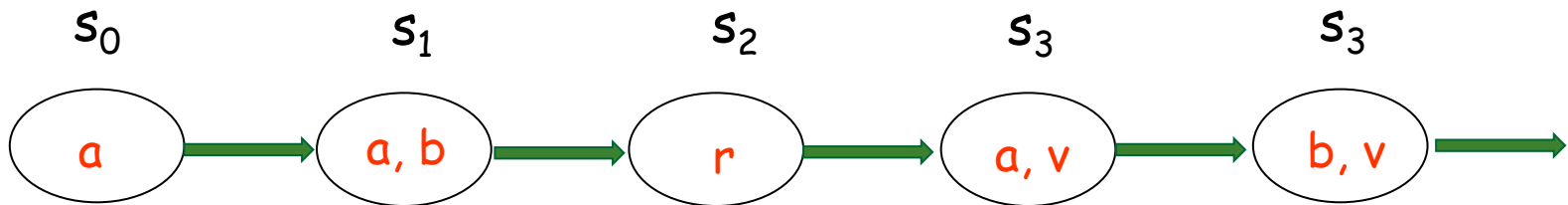
$Fr \wedge G(r \rightarrow XG\neg r)$

- $B, s_0 \models Fr \wedge G(r \rightarrow XG\neg r)$
- $B, s_1 \models Fr \wedge G(r \rightarrow XG\neg r)$
- $B, s_3 \models Fr \wedge G(r \rightarrow XG\neg r)$

Example: $G(a \rightarrow F b)$

- "Whenever a happens, b must eventually happen."
- $G(a \rightarrow F b)$

$G(a \rightarrow F b)$



- $B, s_0 \models ?$
- $B, s_1 \models ?$
- $B, s_3 \models ?$

Syntactic or Semantic Equivalence?

- $(A+B)' \equiv (A+B)' = A' \bullet B'$
- $(A+B)' = A' \bullet B' \iff \forall I. ([(A+B)']_I = [A' \bullet B']_I)$
- $A \neq B \iff \exists I. ([A]_I \neq [B]_I)$

Equivalence Among Formulas

- G and F are dual:

- $Gp = \neg F \neg p$

$$Fp = \neg G \neg p$$

- $G\neg p = \neg Fp$

$$F\neg p = \neg Gp,$$

- $X\neg p = \neg Xp$

- $Gp \wedge Gq = G(p \wedge q)$

$$Fp \wedge Fq \neq F(p \wedge q)$$

- $Fp \vee Fq = F(p \vee q)$

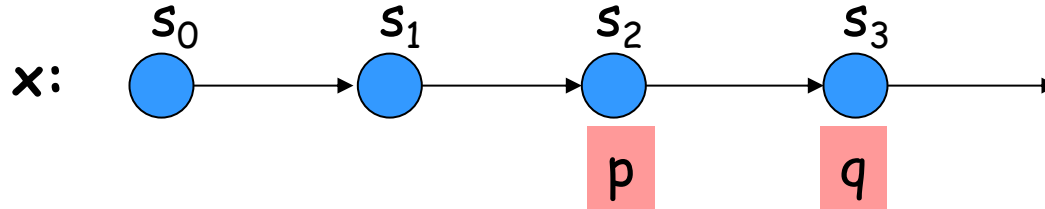
$$Gp \vee Gq \neq G(p \vee q)$$

Prove an Equation: $Fp \vee Fq = F(p \vee q)$

- Part 1: $\Rightarrow$
 - $\Rightarrow$ From the semantics, $B, s_0 \models Fp \vee Fq$
 - $\Rightarrow B, s_0 \models Fp$ or $B, s_0 \models Fq$
 - $\Rightarrow (\exists j. s_j \models p)$ or $(\exists k. s_k \models q)$
 - $\Rightarrow (\exists h. s_h \models p \vee q)$
 - $\Rightarrow B, s_0 \models F(p \vee q)$
- Part 2: $\Leftarrow$
 - $\Rightarrow$ From the semantics, $B, s_0 \models F(p \vee q)$
 - $\Rightarrow (\exists j. s_j \models p \vee q)$
 - $\Rightarrow (\exists j. s_j \models p \text{ or } s_j \models q)$
 - $\Rightarrow (\exists j. s_j \models p)$ or $(\exists j. s_j \models q)$
 - $\Rightarrow B, s_0 \models Fp$ or $B, s_0 \models Fq$
 - $\Rightarrow B, s_0 \models Fp \vee Fq$

- B can be any interpretation.
- i.e., for any B .

Disprove an Equation: $Fp \wedge Fq = F(p \wedge q)$



- Construct an interpretation B such that
- $B, s_0 \models (Fp \wedge Fq)$, but
- $B, s_0 \not\models F(p \wedge q)$,

References

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